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$$1) \quad \exists! A (F \subseteq P(A) \wedge \forall B (F \subseteq P(B) \rightarrow A \subseteq B))$$

$$\text{let } A = UF$$

We will just prove that $F \subseteq P(A) = P(UF)$

Suppose $C \in F$ is arbitrary

$$C \in P(UF)$$

$$C \subseteq UF$$

$$y \in C$$

$$\forall y [y \in C \rightarrow y \in UF]$$

Clearly $D = C$ as $C \in F$ & $y \in C$ is
the right choice

$$\exists D (D \in F \wedge y \in D)$$

Now

$$\forall B (F \subseteq P(B) \rightarrow A \subseteq B)$$

let B be arbitrary &

$$A \subseteq B$$

$$F \subseteq P(B)$$

$$UF \subseteq B$$

$$\forall C [C \in F \rightarrow C \subseteq B] \text{ --- (1)}$$

$$\forall x [x \in UF \rightarrow x \in B]$$

Supp $x \in UF$

$$x \in B$$

$$\exists D \in F (x \in D)$$

Since $D \in F$ so by (1) $D \subseteq B$ & since $x \in D$ so $x \in B$.

Now we have to prove that $A = UF$ is unique. Suppose
there are A_1 & A_2 that satisfy these 2 properties i.e.

$$F \subseteq P(A_1) \text{ --- (1)}$$

$$F \subseteq P(A_2) \text{ --- (2)}$$

$$\forall B [F \subseteq P(B) \rightarrow A_1 \subseteq B] \text{ --- (3)}$$

$$\forall B [F \subseteq P(B) \rightarrow A_2 \subseteq B] \text{ --- (4)}$$

putting (1) in (4) we get $A_2 \subseteq A_1$ & putting (2) in (3)

we get $A_1 \subseteq A_2$ so $A_1 = A_2$

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$$2) \Rightarrow \exists! m \in \mathbb{R}^+ \left[\forall x \in \mathbb{R}^+ \left(\frac{x}{x+1} < m \right) \wedge \forall y \in \mathbb{R}^+ \left(\forall x \in \mathbb{R}^+ \left(\frac{x}{x+1} < y \right) \right) \rightarrow m \leq y \right]$$

Let $m=1$

As $\frac{x}{x+1} = \frac{x+1-1}{x+1} = 1 - \left(\frac{1}{x+1} \right) < 1$ for every $x \in \mathbb{R}^+$ because

$\frac{1}{x+1} > 0$ for all $x \in \mathbb{R}^+$.

But an easy way to prove this is as $x < x+1$ for every $x \in \mathbb{R}^+$ so $\frac{x}{x+1} < 1$ so $m=1$.

To prove next property suppose $y \in \mathbb{R}^+$

Suppose $x \in \mathbb{R}^+$ & $\forall x \left(\frac{x}{x+1} < y \right)$

$$m \leq y$$

we will prove by contradiction

$$1 \leq y$$

suppose $y < 1$

Let $x = \frac{y}{1-y}$ AS $y \in \mathbb{R}^+$ so $0 < y < 1$

so $x = \frac{y}{1-y}$ is positive real

putting $x = \frac{y}{1-y}$ in $\frac{x}{x+1}$ gives $\frac{\frac{y}{1-y}}{\frac{y}{1-y} + 1} = \frac{y}{1-y+y} = \frac{y}{1} = y$

= y but this is contradiction

$$\frac{\frac{y}{1-y} + 1}{1} = \frac{y+1-y}{1-y} = \frac{1}{1-y}$$

as $\frac{x}{x+1} < y$ so $y \geq 1$.

To see that m is unique. Suppose m_1 & m_2 satisfy these properties that is

$$\forall x \left(\frac{x}{x+1} < m_1 \right) \quad (1)$$

$$\forall x \left(\frac{x}{x+1} < m_2 \right) \quad (2)$$

putting (1) in (2) we get

$$m_2 \leq m_1$$

& (2) in (1) we get

$$m_1 \leq m_2$$

so $m_1 = m_2$

$$\forall y \in \mathbb{R}^+ \left(\forall x \left(\frac{x}{x+1} < y \right) \rightarrow m_1 \leq y \right) \quad (3)$$

$$\forall y \in \mathbb{R}^+ \left(\forall x \left(\frac{x}{x+1} < y \right) \rightarrow m_2 \leq y \right) \quad (4)$$

3) $P(A|B) \setminus (P(A) \setminus P(B)) = \{\emptyset\}$
 $\rightarrow x \in P(A|B) \setminus (P(A) \setminus P(B)) \quad x \in \{\emptyset\}$
 $x \in P(A|B) \wedge \neg x \in P(A) \setminus P(B) \quad x = \emptyset$
 $x \subseteq A \setminus B \wedge x \not\subseteq P(A) \vee x \subseteq P(B)$
 $\bullet x \in A|B = \forall y [y \in x \rightarrow (y \in A \wedge y \notin B)] \quad (a)$
 $x \not\subseteq P(A) \vee x \subseteq P(B)$
 $= \neg(x \subseteq A) \vee x \subseteq B = \exists y [y \in x \wedge y \notin A] \vee \forall y [y \in x \rightarrow y \in B]$
 Suppose $x \neq \emptyset \rightarrow$ ① Contradiction ②
 Case 1) $\exists y [y \in x \wedge y \notin A]$
 as $x \neq \emptyset$ so $y \in x \wedge y_0 \notin A$. Since $y_0 \in x$ so $y_0 \in (a)$
 $y_0 \in A \wedge y_0 \notin B$. But this is contradiction over $y_0 \notin A$.
 Case 2) $\forall y [y \in x \rightarrow y \in B]$. Since $x \neq \emptyset$ so $\exists a (a \in x)$ since
 $a \in x$ so by (a) $a \in A \wedge a \notin B$ & by ② $a \in B$ but this
 is contradiction

$\leftarrow$ Suppose $x \in \{\emptyset\}$ so $x = \emptyset$ $P(A|B) \setminus (P(A) \setminus P(B))$
 Subst $x = \emptyset$ so $\forall y (y \notin x)$ as x is empty
 Now $P(A|B) \setminus (P(A) \setminus P(B))$ means
 $\forall y [y \in x \rightarrow (y \in A \wedge y \notin B)] \quad \text{①}$
 $\& \exists y [y \in x \wedge y \notin A] \vee \forall y [y \in x \rightarrow y \in B] \quad \text{②}$
 Now as $y \in x$ is false for every y so ① & second part
 of ② are vacuously True as False $\rightarrow$ Anything is True.
 • Note that we can't conclude $y \in A$ if we have e.g. $y \notin x \rightarrow y \in A$
 & we know that $y \notin x$. We can only conclude that implication
 $y \notin x \rightarrow y \in A$ is True.

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Another more good approach is to use the fact that $\emptyset \subseteq A$ for every set A . $\forall A (\emptyset \subseteq A)$. so $\emptyset \subseteq B$. & $\emptyset \in P(B)$ so $\emptyset \notin P(A) \setminus P(B)$ & since $\emptyset \in P(A \cap B)$ (As $\emptyset \subseteq$ every set) so $\emptyset \in P(A \cap B) \setminus (P(A) \setminus P(B))$ so $\emptyset \in C$.

4) (a) $\rightarrow$ (b)

$$\text{suppose } (A^3 C) \cap (B^3 C) = \emptyset$$

$$A \cap B \subseteq C$$

$$C \subseteq A \cup B$$

$$\text{suppose } \exists x [x \in A^3 C \wedge x \in B^3 C]$$

$$\forall x [(x \notin A \vee x \in C) \wedge (x \notin C \vee x \in A)] \vee [(x \notin B \vee x \in C) \wedge (x \notin C \vee x \in B)] \quad \text{--- (1)}$$

$$\forall x [x \in A \wedge x \in B \rightarrow x \in C]$$

$$\forall x [x \in C \rightarrow x \in A \vee x \in B]$$

$$\text{suppose } x \in A \wedge x \in B \rightarrow x \in C$$

$$\text{Since } x \in A \text{ so } x \in C. \text{ Case 1) } (x \notin A \vee x \in C) \wedge (x \notin C \vee x \in A)$$

$$\text{Since } x \in A \text{ so } x \in C.$$

$$\text{Case 2) } (x \notin B \vee x \in C) \wedge (x \notin C \vee x \in B). \text{ Since } x \in B \text{ so } x \in C. \text{ so } x \in A \cup B.$$

(b) $\rightarrow$ (c)

$$\text{suppose } x \in C \rightarrow x \in A \vee x \in B.$$

$$(A \cap B) \subseteq C$$

$$\text{Case 1) } (x \notin A \vee x \in C) \wedge (x \notin C \vee x \in A)$$

$$\text{Since } x \in C \text{ so } x \in A. \text{ so } x \in A \vee x \in B \text{ is true.}$$

$$\text{Case 2) } (x \notin B \vee x \in C) \wedge (x \notin C \vee x \in B). \text{ Since } x \in C$$

$$\text{so } x \in B \text{ so } x \in A \vee x \in B \text{ then } x \in A \cup B$$

(b) $\rightarrow$ (c)

$$A^3 C \subseteq A^3 B$$

$$A \cap B \subseteq C$$

$$C \subseteq A \cup B$$

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$$\forall x [x \in A \wedge x \in B \rightarrow x \in C] \quad \text{--- (1)}$$

$$\forall x [x \in C \rightarrow x \in A \vee x \in B] \quad \text{--- (2)}$$

Suppose $x \in A^c$

$$(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A)$$

$$(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)$$

Case 1) Suppose $x \in A \wedge x \notin C$. Since $x \notin C$ so by (1) $x \notin A \vee x \notin B$.
But since $x \in A$ so $x \notin B$. So $x \in A \setminus B$. (Proves 1st case of goal.)

Case 2) Suppose $x \in C \wedge x \notin A$. Since $x \in C$ so $x \in A \vee x \in B$ by (2).
But as $x \notin A$ so $x \in B$. So $x \in B \setminus A$. (2nd case of goal)

$$(C) \rightarrow (A)$$

$$A^c \subseteq A^c \cup B$$

$$(A^c) \cap (B^c) = \emptyset$$

$$\forall x [x \in A^c \rightarrow x \in A^c \cup B]$$

$$\forall x [(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A) \rightarrow (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)] \quad \text{--- (3)}$$

$$\forall x [x \in A^c \rightarrow x \notin B^c]$$

$$\text{Suppose } x \in A^c$$

$$x \notin B^c$$

$$(x \in A \wedge x \notin C) \vee (x \in C \wedge x \notin A) \quad \sim [(x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)]$$

Since $x \in A^c$ we can conclude

$$(x \notin B \vee x \in C) \wedge (x \notin C \vee x \in B)$$

by (1) that $(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)$

Case 1) $\rightarrow$ 1st of goal follows. $x \notin B$

Case 2) $\rightarrow$ 2nd of goal follows. $x \in B$

*) $\{A_i \mid i \in I\}$ is a family

$$P(\bigcup_{i \in I} A_i) \subseteq \bigcup_{i \in I} P(A_i) \quad \exists i \in I \forall j \in I (A_j \subseteq A_i)$$

$$\forall x [x \in P(\bigcup_{i \in I} A_i) \rightarrow x \subseteq \bigcup_{i \in I} A_i] \quad (1)$$

$$\forall x [x \subseteq \bigcup_{i \in I} A_i \rightarrow \exists i \in I (x \subseteq A_i)] \quad (2)$$

If we attempt some examples, we will see that a set with all $A_i \subseteq A_i$ is $\bigcup_{i \in I} A_i$

Now clearly $\bigcup_{i \in I} A_i \subseteq \bigcup_{i \in I} A_i$ (every set is subset of itself)

$$\text{So } \bigcup_{i \in I} A_i \in P(\bigcup_{i \in I} A_i)$$

$$\text{So by (1) } \bigcup_{i \in I} A_i \in \bigcup_{i \in I} P(A_i) \quad \text{put this in (2) directly}$$

Here is crucial part $\bigcup_{i \in I} A_i = \{x \mid \exists i \in I (x \subseteq A_i)\}$

$$\& \bigcup_{i \in I} P(A_i) = \{x \mid \exists i \in I (x \subseteq A_i)\} \quad \begin{array}{l} \text{elements of } A_i \\ \text{subsets of } A_i \end{array}$$

So $\bigcup_{i \in I} A_i$ belongs to subsets of A_i

$$\text{So } \bigcup_{i \in I} A_i \subseteq A_i \quad (2) \quad (\text{Because we took } A_i = \bigcup_{i \in I} A_i)$$

So there does exist so it is not that (2) holds

Now suppose $j \in I$ be arbitrary, we know that

$$A_j \subseteq \bigcup_{i \in I} A_i \quad (\text{Previously proved here, } j \text{ also } \in I \text{ so this kind of union } A_i \subseteq \bigcup_{i \in I} A_i \text{ which is a fact})$$

$$\text{So } A_j \subseteq A_i \quad (\because A \subseteq B \& B \subseteq C \text{ then } A \subseteq C)$$

6)

$$(a) \quad I = \{ \emptyset \}$$

$$J = \{ \emptyset \}$$

$$J \neq \emptyset$$

$$I \neq \emptyset$$

$$\forall x (x \in I \rightarrow x \neq \emptyset)$$

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$$\bigcup_{i \in I} A_i = \bigcup_{x \in F} (\bigcup_{i \in I} A_i)$$

→ Supp. $x \in \bigcup_{i \in I} A_i$

$$\exists i \in I (x \in A_i)$$

$$i_0 \in I$$

$$x \in A_{i_0}$$

$$x \in \bigcup_{x \in F} (\bigcup_{i \in I} A_i)$$

$$\exists x \in F (x \in \bigcup_{i \in I} A_i)$$

$$\exists x \in F (\exists i \in I (x \in A_i))$$

Since $i_0 \in I$ & also $I = \bigcup F$

$$\text{so } i_0 \in \bigcup F \rightarrow \exists x \in F (i_0 \in x)$$

$$x_0 \in F$$

$$i_0 \in x_0 \text{ so we find } x_0 \in F \text{ } i_0 \in x \text{ \& } x \in A_{i_0}$$

← Supp. $\exists x \in F (\exists i \in I (x \in A_i))$

$$x_0 \in F \text{ --- (1)}$$

$$i_0 \in x_0 \text{ --- (2)}$$

$$x_0 \in A_{i_0}$$

$$\exists x \in F (x \in \bigcup_{i \in I} A_i)$$

$$\exists i \in I (x_0 \in A_i)$$

by (1) & (2) $i_0 \in \bigcup F$

Since $\bigcup F = I$ so $i_0 \in I$

also $x_0 \in A_{i_0}$ so $\exists i \in I (x_0 \in A_i)$

$$(b) \bigcap_{i \in I} A_i = \bigcap_{x \in F} (\bigcap_{i \in I} A_i)$$

Supp. $x \in \bigcap_{i \in I} A_i$

$$\forall i (i \in I \rightarrow x \in A_i) \text{ --- (1)}$$

Suppose

$x \in F$ & if x is arbitrary

$$x \in \bigcap_{x \in F} (\bigcap_{i \in I} A_i)$$

$$\forall x \in F (x \in \bigcap_{i \in I} A_i)$$

$$\forall x \in F (\forall i \in I (x \in A_i))$$

so $i \in \bigcup F$

Since $\bigcup F = I$ so $i \in I$

& from (1) since $i \in I$ so $x \in A_i$

$$\leftarrow \text{Supp } \forall x \in F (\forall i \in X (x \in A_i)) \quad \forall i (i \in I \rightarrow x \in A_i)$$

$$\forall x [x \in F \rightarrow \forall i \in X (x \in A_i)] \quad \text{let } i \in X \text{ arbitrary}$$

$$\text{Supp } x \in F \quad x \in A_i$$

Since $I = U F$

$$\text{so } i \in U F \rightarrow \exists x \in F (i \in x)$$

$$\text{so } x_0 \in F (i \in x_0)$$

Since $x_0 \in F$ so by ① $\forall i \in x_0 (x_0 \in A_i)$ so $x_0 \in A_i$

$$(C) \bigcup_{i \in J} A_i \subseteq \bigcap_{x \in F} (\bigcup_{i \in x} A_i)$$

Supp $x \in \bigcup_{i \in J} A_i$

$$\exists i \in J (x \in A_i)$$

so $i_0 \in J (x \in A_{i_0})$

Supp $x \in F$

Since $i_0 \in J$ & $J = \bigcap F$

so $i_0 \in \bigcap F$

which means $\forall x \in F (i_0 \in x)$

Since $x \in F$ so $i_0 \in x$

so we found an $i = i_0$ such that $i_0 \in x$ & $x \in A_{i_0}$.

② (C) (part 2)

$$\bigcup_{i \in J} A_i = \bigcap_{x \in F} (\bigcup_{i \in x} A_i) ?$$

Supp $A_i = \{i, i+5, i+6\}$

$$F = \{ \{1, 2\}, \{1, 3\} \}$$

Then $J = \bigcap F = \{1\}$

$I = U F = \{1, 2, 3\}$

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Now $\bigcup_{i \in S} A_i = A_1 = \{6, 7\} \rightarrow (\because \{1+5, 1+6\})$

But

$$\begin{aligned} \bigcap_{x \in S} (A_i) &= (\{6, 7\} \cup \{7, 8\}) \cap (\{6, 7\} \cup \{8, 9\}) \\ &= \{6, 7, 8\} \cap \{6, 7, 8, 9\} \\ &= \{6, 7, 8\} \end{aligned}$$

so these are not equal. Notice we found counterexample by translating into logical symbols & try to proof but we have problem that it's $\neq$ but no guarantee that it's $= \cap$ or $\neq$.
that helped us construct counterexample

(d) $\bigcap_{i \in S} A_i \subseteq \bigcup_{x \in F} (\bigcap_{i \in S} A_i)$

supp $x \in \bigcup_{x \in F} (\bigcap_{i \in S} A_i)$	$x \in \bigcap_{i \in S} A_i$
$\exists x \in F (x \in \bigcap_{i \in S} A_i)$	$\forall i \in S (x \in A_i)$
$\exists x \in F (\forall i \in S (x \in A_i))$ ①	$x \in A_i$

suppose it's be arbitrary
initially for ①, we get
 $x_0 \in F (\forall i \in S (x_0 \in A_i))$ ②
as it's so it's $\neq$ because $S = \emptyset$

so $\forall x (x \in F \rightarrow i \in x)$
as $x_0 \in F$ so $i \in x_0$ (from ②)
& as $i \in x_0$ so $x_0 \in A_i$